

# Delhi Public School

Subject: Physics | Grade: 11th | Class: 11th

Time: 2 Hours

Maximum Marks: 16

Name: \_\_\_\_\_

Roll No: \_\_\_\_\_

Section: \_\_\_\_\_

## General Instructions:

1. Read all questions carefully before answering.
2. Write neatly and legibly.

## Section A

This section contains 5 multiple-choice questions. Each question carries 1 mark. Select the single correct option. Time Allowed: 1 hour 30 minutes (Total Exam).

**Q1.** The dimensional formula of Universal Gravitational Constant (G) is: [Easy] [1]

- a.  $[M^{-1} L^3 T^{-2}]$
- b.  $[M L^3 T^{-2}]$
- c.  $[M^{-1} L^2 T^{-2}]$
- d.  $[M^{-2} L^3 T^{-1}]$

**Expected Answer:**  $[M^{-1} L^3 T^{-2}]$

**Q2.** A vehicle travels the first half of distance L with speed  $v_1$  and the remaining half distance with speed  $v_2$ . The average speed of the vehicle is: [Easy] [1]

- a.  $(v_1 + v_2) / 2$
- b.  $(2 * v_1 * v_2) / (v_1 + v_2)$
- c.  $(v_1 * v_2) / (v_1 + v_2)$
- d.  $\sqrt{v_1 * v_2}$

**Expected Answer:**  $(2 * v_1 * v_2) / (v_1 + v_2)$

**Q3.** At what angle of projection with the horizontal will the maximum height of a projectile be equal to its horizontal range? [Moderate] [1]

- a.  $\theta = \tan^{-1}(1)$
- b.  $\theta = \tan^{-1}(2)$
- c.  $\theta = \tan^{-1}(4)$
- d.  $\theta = 45$  degrees

**Expected Answer:**  $\theta = \tan^{-1}(4)$

**Q4.** The work done by a conservative force along a closed path/loop is always: [Moderate] [1]

- a. positive
- b. negative
- c. zero
- d. dependent on the nature of the force

**Expected Answer:** zero

**Q5.** If the kinetic energy of a body is increased by 300%, what is the percentage increase in its momentum? [Hard] [1]

- a. 50%
- b. 100%
- c. 150%
- d. 200%

**Expected Answer:** 100%

## Section B

This section contains 3 short-answer questions. Each question carries 2 marks. Write precise answers.

**Q6.** Explain why rain drops fall with a constant terminal velocity instead of accelerating continuously under gravity. [Easy] [2]

**Expected Answer:** As a rain drop falls, it experiences downward force of gravity and upward viscous drag plus upthrust. As velocity increases, the viscous force ( $F = 6 \pi \eta r v$ ) increases. Eventually, the upward forces equal the downward force of gravity, resulting in a net force of zero. Consequently, the acceleration becomes zero and the drop falls with a constant terminal velocity.

**Q7.** A block of mass 2 kg is placed on a horizontal surface. If the coefficient of static friction between the block and the surface is 0.4, find the frictional force acting on the block when a horizontal force of 5 N is applied to it. (Take  $g = 10 \text{ m/s}^2$ ) [Moderate] [2]

**Expected Answer:** The maximum limiting static friction is  $f_{\text{max}} = \mu_s \cdot R = \mu_s \cdot m \cdot g = 0.4 \cdot 2 \cdot 10 = 8 \text{ N}$ . Since the applied horizontal force (5 N) is less than the limiting friction (8 N), the block does not move. Therefore, static friction adjusts itself to equal the applied force. The frictional force acting on the block is 5 N.

**Q8.** State Kepler's second law of planetary motion (Law of Areas) and show that it is a direct consequence of the conservation of angular momentum. [Hard] [2]

**Expected Answer:** Kepler's second law states that the line joining a planet to the sun sweeps out equal areas in equal intervals of time (i.e., areal velocity  $dA/dt$  is constant). Mathematically,  $dA/dt = L / (2 \cdot m)$ , where  $L$  is the angular momentum and  $m$  is the mass of the planet. Since the gravitational force is central, the external torque acting on the planet is zero, which means its angular momentum  $L$  is conserved (constant). Therefore,  $dA/dt$  remains constant.

### Section C

*This section contains 5 long-answer questions. Each question carries 1 mark. Write structured derivations and explanations as required.*

**Q9.** Derive the mathematical expression for the variation of acceleration due to gravity ( $g$ ) with depth ( $d$ ) below the surface of the Earth. [Moderate] [1]

**Expected Answer:** Let the Earth be a uniform sphere of radius  $R$  and mass  $M$ . At the surface,  $g = G \cdot M / R^2 = 4/3 \pi \cdot G \cdot R \cdot \rho$ . At depth  $d$ , the acceleration due to gravity is  $g_d = G \cdot M' / (R-d)^2$ , where  $M'$  is the mass of the inner sphere of radius  $(R-d)$ . Thus,  $g_d = 4/3 \pi \cdot G \cdot (R-d) \cdot \rho$ . Dividing  $g_d$  by  $g$  gives:  $g_d / g = (R-d)/R = 1 - d/R$ . Hence,  $g_d = g \cdot (1 - d/R)$ .

**Q10.** State and prove Bernoulli's principle for the flow of an ideal, non-viscous, incompressible fluid in a streamlined flow. [Moderate] [1]

**Expected Answer:** Statement: For an incompressible, non-viscous fluid undergoing steady flow, the sum of pressure energy, kinetic energy, and potential energy per unit volume remains constant along a streamline:  $P + 1/2 \cdot \rho \cdot v^2 + \rho \cdot g \cdot h = \text{Constant}$ . Proof: Consider a pipe of varying cross-section. The work done on the fluid by pressure differences equals the change in its kinetic and potential energies (Work-Energy Theorem).  $W = (P_1 - P_2) \cdot dV$ . Change in KE =  $1/2 \cdot dm \cdot (v_2^2 - v_1^2)$ . Change in PE =  $dm \cdot g \cdot (h_2 - h_1)$ . Equating  $W = dKE + dPE$  and dividing by  $dV$  (since  $dm/dV = \rho$ ) yields:  $P_1 - P_2 = 1/2 \cdot \rho \cdot (v_2^2 - v_1^2) + \rho \cdot g \cdot (h_2 - h_1)$ . Rearranging gives  $P_1 + 1/2 \cdot \rho \cdot v_1^2 + \rho \cdot g \cdot h_1 = P_2 + 1/2 \cdot \rho \cdot v_2^2 + \rho \cdot g \cdot h_2$ .

**Q11.** State the law of equipartition of energy. Using this law, calculate the molar specific heat capacities at constant volume ( $C_v$ ) and constant pressure ( $C_p$ ) for a rigid diatomic gas. [Easy] [2]

**Expected Answer:** Law of Equipartition of Energy: For any system in thermal equilibrium, the total energy is equally distributed among its various degrees of freedom, and the energy associated with each degree of freedom per molecule is  $1/2 \cdot k \cdot T$ . A rigid diatomic gas has 5 degrees of freedom (3 translational, 2 rotational). Internal energy per mole  $U = 5 \cdot (1/2 \cdot R \cdot T) = 5/2 \cdot R \cdot T$ . Molar heat capacity at constant volume  $C_v = dU/dT = 5/2 \cdot R$ . Using Mayer's relation,  $C_p = C_v + R = 5/2 \cdot R + R = 7/2 \cdot R$ .

**Q12.** What is a simple pendulum? Derive an expression for its time period and show that for small angular displacements, its motion is Simple Harmonic Motion (SHM). [Moderate] [1]

**Expected Answer:** A simple pendulum consists of a heavy point mass (bob) suspended by a weightless, inextensible string from a rigid support. When pulled aside by a small angle  $\theta$ , the restoring torque is  $\tau = -m \cdot g \cdot l \cdot \sin(\theta)$ . For small angles,  $\sin(\theta)$  approximately equals  $\theta$  (in radians). Thus,  $\tau = -m \cdot g \cdot l \cdot \theta$ . Also,  $\tau = I \cdot \alpha = m \cdot l^2 \cdot \alpha$ . Equating these,  $m \cdot l^2 \cdot \alpha = -m \cdot g \cdot l \cdot \theta$ , which simplifies to  $\alpha = -(g/l) \cdot \theta$ . Since angular acceleration ( $\alpha$ ) is directly proportional to displacement ( $\theta$ ) and directed opposite to it, the motion is SHM. Comparing with  $\alpha = -\omega^2 \cdot \theta$ , we get  $\omega = \sqrt{g/l}$ . The time period is  $T = 2 \cdot \pi / \omega = 2 \cdot \pi \cdot \sqrt{l/g}$ .

**Q13.** Discuss the formation of standing waves in a stretched string of length  $L$  fixed at both ends. Obtain the general mathematical equation for standing waves and derive expressions for the frequencies of the first three harmonics. [Hard] [1]

**Expected Answer:** When two identical progressive waves travel in opposite directions along a stretched string, they superpose to form standing waves. Let  $y_1 = A \cdot \sin(k \cdot x - \omega \cdot t)$  and  $y_2 = -A \cdot \sin(k \cdot x + \omega \cdot t)$  (reflected from the boundary). Superposition yields:  $y = y_1 + y_2 = -2 \cdot A \cdot \sin(k \cdot x) \cdot \cos(\omega \cdot t)$ . Boundary conditions demand that at  $x = 0$  and  $x = L$ , displacement  $y = 0$ . Thus,  $\sin(k \cdot L) = 0$ , implying  $k \cdot L = n \cdot \pi$  where  $n = 1, 2, 3, \dots$ . Since  $k = 2 \cdot \pi / \lambda$ , we get  $\lambda_n = 2 \cdot L / n$ . The frequencies are given by  $f_n = v / \lambda_n = n \cdot v / (2 \cdot L)$ . 1. First Harmonic ( $n=1$ , fundamental):  $f_1 = v / (2 \cdot L)$ . 2. Second Harmonic ( $n=2$ , 1st overtone):  $f_2 = 2 \cdot v / (2 \cdot L) = 2 \cdot f_1$ . 3. Third Harmonic ( $n=3$ , 2nd overtone):  $f_3 = 3 \cdot v / (2 \cdot L) = 3 \cdot f_1$ .

— End of Question Paper —